

Customer Retention & Churn Analysis

Task 2 · Data Science & Analytics Internship · Future Interns

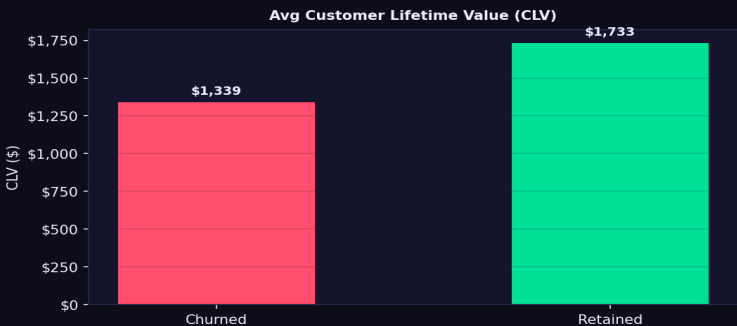
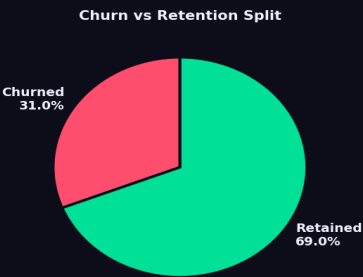
Dataset: Telco Customer Churn (Simulated) · Tools: Python · Pandas · Matplotlib · Seaborn · Repo: FUTURE_DS_02

Executive Summary

This analysis examines 7,043 telecom customers to identify churn drivers, retention patterns, and customer lifetime value differences between churned and retained segments. The overall churn rate stands at 31.0%, representing 2,185 lost customers and \$142,607 in monthly recurring revenue at risk. Contract type, internet service type, payment method, and customer tenure are the four strongest predictors of churn. This report delivers segment-level insights and a prioritised action plan to reduce churn and improve retention.

Key Performance Indicators

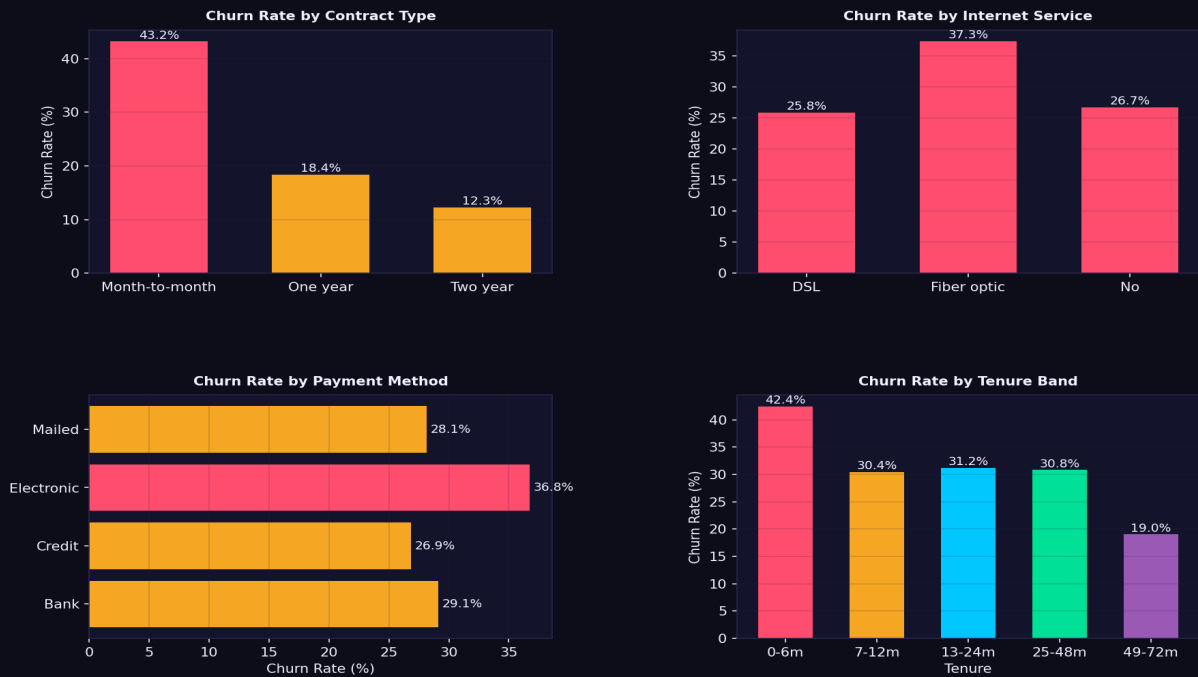
7,043 Total Customers	2,185 Churned Customers	31.0% Churn Rate
\$142,607 Monthly Rev at Risk	\$1,733 Avg CLV (Active)	28.8m Avg Tenure (Active)



Retained customers generate \$1.3x more lifetime value than churned customers — making retention investment directly tied to revenue growth.

Churn Driver Analysis

Four factors dominate churn behaviour: contract type, internet service, payment method, and tenure. Each segment shows a materially different churn rate.

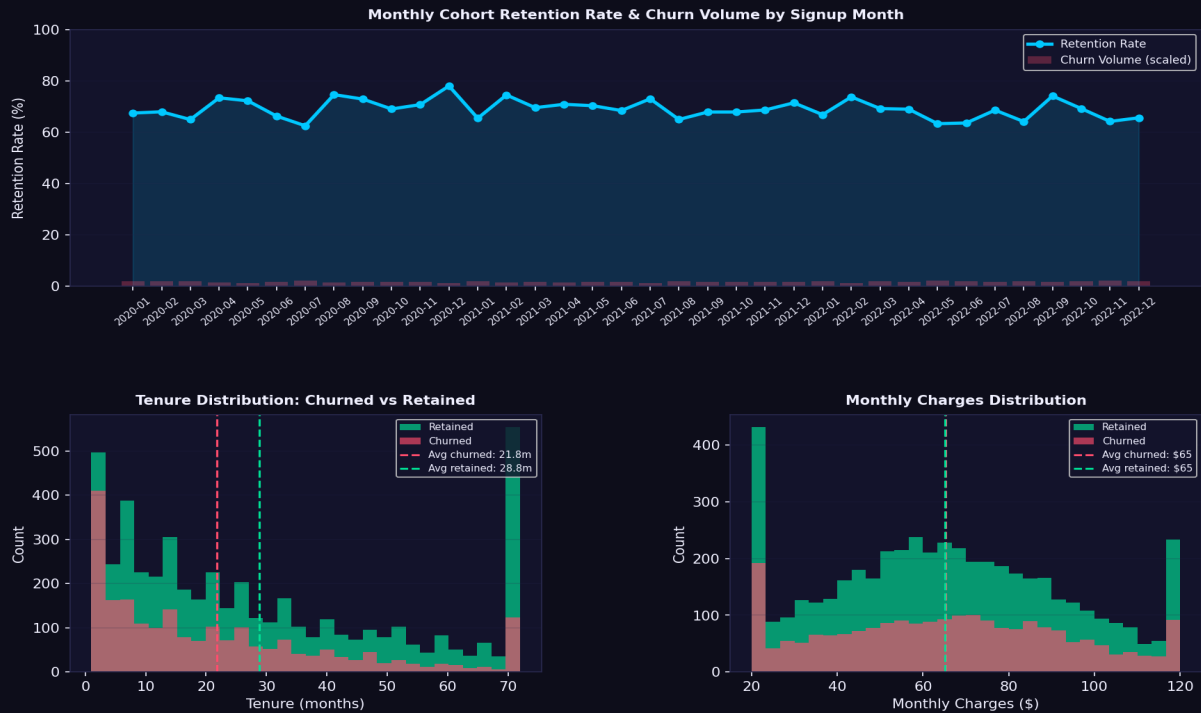


Churn Driver Summary Table

Factor	Segment	Churn Rate	Risk Level	Priority
Contract	Month-to-month	43.2%	■ Critical	#1
Contract	Two year	12.3%	■ Low	#5
Internet	Fiber optic	37.3%	■ High	#2
Payment	Electronic check	36.8%	■ Medium	#3
Tenure	0–6 months	42.4%	■ Critical	#1
Tenure	49–72 months	19.0%	■ Low	#5

Cohort Retention & Customer Lifetime Analysis

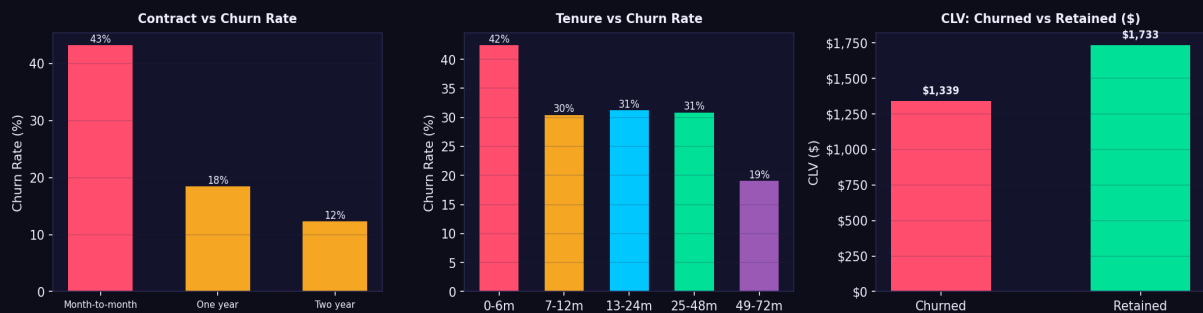
Cohort analysis tracks retention by the month customers first signed up. Tenure and monthly charges distributions reveal clear separation between churned and retained segments.



Churned customers leave after an average of 21.8 months vs 28.8 months for retained customers — a 7.0 month gap. Churned customers also pay \$65/month on average vs \$65 for retained — suggesting price sensitivity in high-charge segments.

Key Business Insights

- **Contract Lock-in Is the #1 Retention Driver:** Month-to-month customers churn at 43.2% vs just 12.3% for two-year contracts. This 30.9pp gap is the single largest controllable churn lever in the business.
- **First 6 Months Are the Critical Risk Window:** Customers in their first 6 months churn at 42.4% — the highest of any tenure band. By month 49+, churn drops to just 19.0%. Early engagement programs are non-negotiable.
- **Fiber Optic Quality May Be Driving Dissatisfaction:** Fiber optic customers churn at 37.3% vs 25.8% for DSL. Given Fiber is the premium product, this suggests a service quality or expectation mismatch.
- **Electronic Check Payment Correlates With High Churn:** Electronic check users churn at 36.8% — the highest payment method churn. This segment may include price-sensitive or less-engaged customers who haven't committed to auto-pay.
- **CLV Gap Is the True Business Impact:** Retained customers generate \$1,733 avg CLV vs \$1,339 for churned — a 1.3x difference. Every percentage point reduction in churn meaningfully impacts revenue.
- **Monthly Revenue Risk Is Concentrated:** \$142,607/month in MRR is at risk from already-churned customers. Reactivation campaigns targeting recent churners can recover a portion of this revenue.



Actionable Recommendations

- **Incentivize Long-Term Contracts:** Offer 15-20% discount for month-to-month customers who switch to annual plans. Even converting 20% of M2M customers to annual contracts would reduce overall churn by ~6pp.
- **Launch a 90-Day Onboarding Program:** Create a structured onboarding journey for new customers (Day 1, 7, 30, 60, 90 touchpoints). Churn in the 0-6 month window can be cut significantly with proactive engagement.
- **Investigate Fiber Optic Service Quality:** Run NPS surveys specifically for Fiber optic customers. If speed or reliability is the issue, a service quality fix has outsized retention impact.
- **Convert Electronic Check Users to Auto-Pay:** Offer a \$5/month bill credit to customers who switch from electronic check to auto-pay. Auto-pay customers are more engaged and less likely to churn.
- **Build a Churn Early Warning System:** Score customers monthly using tenure + contract + payment method signals. Flag high-risk customers for proactive outreach before they cancel.
- **Reactivation Campaign for Recent Churners:** Target customers who churned in the last 90 days with a win-back offer. Recovering 10% of churned customers would reclaim ~\$14,261/month in MRR.